

Gowtham Sai Kumar Akala

Sr. Network Engineer

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Professional Summary:

Senior Cloud Network & Security Engineer with 5 years of progressive experience advancing from Junior to Senior roles, specializing in the design, deployment, and security of enterprise and multi-cloud infrastructures. Skilled in building scalable **LAN/WAN architectures** using VLANs, STP, BGP, OSPF, MPLS, VRFs, and QoS, with hands-on expertise across Cisco Catalyst, Nexus, ASR, Juniper MX, and Arista platforms.

Proven success in **firewall migrations and security architecture** with Palo Alto NGFW (Panorama/Expedition), Cisco ASA/FTD, Check Point, and Fortinet, implementing **Zero Trust, SASE, and NAC solutions** (Cisco ISE, Aruba ClearPass, ForeScout) to strengthen enterprise security. Designed and supported **Cisco Viptela SD-WAN** rollouts across 40+ sites and **Cisco ACI fabrics** with VXLAN EVPN for microsegmentation and application-centric policies.

Extensive **cloud experience** with AWS (VPC, Transit Gateway, Direct Connect), Azure (ExpressRoute, vWAN, NSGs), and GCP (Interconnect, VPN), including **compliance-aligned security** (PCI DSS, HIPAA, GDPR, NIST 800-53). Strong in **automation and Infrastructure-as-Code**, using Python, Ansible, Terraform, and GitLab CI/CD to accelerate deployments and enforce compliance-as-code.

Hands-on expertise with **load balancers (F5 BIG-IP LTM/GTM/APM/ASM, Citrix NetScaler)**, **wireless (Cisco WLC, Meraki, Aruba, Wi-Fi 6)**, **VPNs (IPsec, SSL, DMVPN, AnyConnect, GlobalProtect with MFA)**, and **observability tools** (Splunk, Grafana, Prometheus, SolarWinds, ThousandEyes). Adept at producing **HLD/LLDs, Visio diagrams, migration runbooks, and compliance dashboards** for enterprise governance.

Recognized for mentoring engineers, leading workshops, collaborating across DevOps/SecOps teams, and driving enterprise initiatives such as **cloud cost optimization, DR/BCP strategies, and compliance governance**. Trusted to deliver secure, resilient, and future-ready infrastructures for complex enterprise environments.

Technical Skills:

Networking Protocols	TCP/IP, VLANs, VTP, STP, RSTP, MSTP, EtherChannel, OSPF, EIGRP, RIP, BGP, MPLS L3VPN, VRFs, QoS, Multicast (PIM, IGMP), Anycast Routing
Routing & Switching Platforms	Cisco Catalyst (ISR, ASR 9000, 9300, 9400, 9500), Cisco Nexus (9k, 7k, 5k, 2k), Juniper MX960, MX Series, Arista Switches
Firewalls & Security	Palo Alto NGFW (Panorama, Expedition, GlobalProtect), Cisco ASA/FTD, Fortinet FortiGate, Check Point Firewalls, Zscaler ZIA/ZPA, Prisma Access
Load Balancers & ADCs	F5 BIG-IP (LTM, GTM, APM, ASM, WAF, iRules), Citrix NetScaler ADC
Cloud Platforms	AWS (VPC, Transit Gateway, Direct Connect, EC2, Route 53, CloudWatch, IAM), Microsoft Azure (vWAN, ExpressRoute, VNets, NSGs, Monitor), Google Cloud (VPC, Interconnect, VPN), Hybrid Multi-Cloud
Cloud Security & Compliance	Zero Trust, SASE, ZTNA, CASB, DLP, IAM Federation (SAML, OAuth2), PKI Lifecycle, HIPAA, PCI-DSS, GDPR, NIST 800-53
Network Access Control (NAC)	Cisco ISE (802.1X, TACACS+, RADIUS, Posture, Profiling), Aruba ClearPass, ForeScout NAC
Wireless & Mobility	Cisco WLC (5508, 9800), Cisco Meraki, Aruba WLAN Controllers, Wi-Fi 6 (802.11ax), WPA3, Ekahau, AirMaget, Cisco Prime
VPN & Remote Access	IPsec VPN, SSL VPN, DMVPN, Cisco AnyConnect, Palo Alto GlobalProtect, GRE Tunnels, MFA (Duo, PingID, Azure MFA)

Automation & Scripting	Python, Ansible, Terraform, GitLab CI/CD, REST APIs, Checkov, tfsec, SonarQube
Container Networking	Kubernetes (CNI plugins), Istio Service Mesh, Microsegmentation
Data Center Technologies	Cisco ACI (Tenants, VRFs, Bridge Domains, Contracts, L3Outs, Service Graphs), VXLAN EVPN Overlays, MPLS TE
Monitoring & Observability	SolarWinds, PRTG, Nagios, Zabbix, ThousandEyes, Splunk, Grafana, Prometheus, FortiAnalyzer, AWS CloudWatch, Azure Monitor
VoIP & Collaboration	Cisco CME, CUCM, SIP, H.323, VoIP QoS
Documentation & ITSM	Visio, ServiceNow CMDB, HLD/LLD Design, Migration Runbooks, Audit Dashboards
Cabling & Infrastructure	Racking/Stacking, Structured Cabling (Cat6, Cat6a, OM3/OM4 Fiber), Patch Panels, TIA/EIA-568 Standards

Education Details:

- Bachelor of Technology – KL University – Electronics and Communicational Engineering, India
- Master of Science – University of Central Missouri - Cybersecurity, Missouri, US

Professional Experience:

Senior Cloud Network & Security Engineer

CITI GROUP JAN 2025 - PRESENT - Kansas, MO

Responsibilities:

- Designed and deployed **hybrid multi-cloud networking** across AWS, Azure, and GCP using Transit Gateway, vWAN, Interconnects, and BGP peering for low-latency workloads.
- Led enterprise rollout of **Cisco Viptela SD-WAN overlay** across 40+ sites, reducing MPLS spend by 50%, improving app performance with SLA-driven policies, and enabling ZTP deployments.
- Architected **Zero Trust Security Framework** combining Zscaler ZIA/ZPA, Palo Alto Prisma Access, Cisco ISE, and ForeScout NAC, enforcing least-privilege, MFA, and posture validation.
- Directed **large-scale firewall migrations** (Cisco ASA, Check Point, Fortinet → Palo Alto NGFW) with Panorama automation, Expedition cleanup, and API-based deployment pipelines.
- Built **Infrastructure-as-Code pipelines** using Terraform, Ansible, GitLab CI/CD, and Python, enabling automated provisioning of VPCs, NSGs, security groups, firewall rules, and route policies.
- Integrated **compliance-as-code** using Checkov, tfsec, and SonarQube, ensuring alignment with **NIST 800-53, HIPAA, GDPR, and PCI DSS**.
- Designed and maintained **ACI microsegmentation** with multi-tenant VRFs, Bridge Domains, Contracts, and L3Out, enforcing east-west traffic policies.
- Managed **F5 BIG-IP LTM/APM/ASM** for application load balancing, SSL VPNs, SSO portals, WAF protection, and traffic optimization using iRules.
- Established centralized **observability and monitoring stack** (Splunk, Grafana, Prometheus, FortiAnalyzer, AWS CloudWatch, Azure Monitor, ThousandEyes), reducing RCA time by 35%.
- Configured and optimized **VPN architectures**: IPsec, SSL, DMVPN, GlobalProtect, integrating MFA (Duo, PingID, Azure MFA) for secure access.
- Performed **capacity planning, DR/BCP design, SLA reporting, and technology roadmaps** for future-proof networking.
- Deployed **SASE architecture** integrating ZTNA, CASB, SWG, and DLP for remote workforce security and SaaS access.

- Mentored junior engineers, conducted workshops on cloud networking, firewall automation, and SD-WAN best practices.
- Supervised vendors during **data center builds**: structured cabling (TIA/EIA-568D), tray layouts, patch panels, and fiber termination.
- Integrated **Aruba ClearPass and Cisco ISE** for advanced NAC, dynamic profiling, and BYOD onboarding.
- Configured **advanced routing features**: BGP route reflectors, Anycast, MPLS TE, VXLAN EVPN overlays, L3VPN segmentation.
- Supported **container networking**: Kubernetes CNI plugins, Istio service mesh, enforcing policy-based east-west traffic in microservices.
- Applied **next-gen monitoring**: ThousandEyes for path tracing, NetFlow/IPFIX for traffic visibility, syslog centralization for audits.
- Created and reviewed **HLDs, LLDs, firewall rule review reports, migration runbooks, audit compliance dashboards**, ensuring governance readiness.
- Partnered with InfoSec to implement **DLP, CASB, encryption standards, IAM federation (SAML, OAuth2), PKI lifecycle management**.
- Drove **cloud cost optimization** by redesigning routing policies, consolidating circuits, and fine-tuning traffic engineering.

Environment: Multi-Cloud (AWS, Azure, GCP), Cisco Viptela SD-WAN, Zero Trust, ZTNA, Prisma Access, Zscaler ZIA/ZPA, Cisco ISE, ForeScout NAC, Palo Alto NGFW, Panorama, Expedition, Terraform, Ansible, GitLab CI/CD, Python, VXLAN EVPN, MPLS TE, F5 LTM, APM, ASM, WAF, Kubernetes, Istio, ThousandEyes, Splunk, Grafana, Prometheus, FortiAnalyzer, PCI DSS, HIPAA, GDPR, NIST 800-53, Compliance-as-Code, SASE, CASB, DLP, PKI.

Network Engineer

AbbVie NOV 2023 - DEC 2024- st Louis, MO

Responsibilities:

- Configured **enterprise routing protocols** (BGP with route reflectors, OSPF area segmentation, MPLS L3VPN, VRFs, EIGRP stubs) across Cisco ASR 9000, Nexus 9k/7k, and Juniper MX960 routers.
- Led migration from legacy MPLS to **Cisco Viptela SD-WAN**, deploying vManage, vBond, vSmart controllers, defining centralized policies, and enabling application-aware routing and SLA-based path selection.
- Designed and implemented **Cisco ACI fabric** (Tenants, VRFs, Bridge Domains, Application Profiles, Contracts, Service Graphs), providing scalable micro segmentation across global data centers.
- Performed **firewall migrations** from Check Point → Palo Alto and ASA → NGFW, validating thousands of rules via **Expedition tool, Panorama templates, and API-driven automation**.
- Configured **F5 BIG-IP LTM/GTM** load balancers for global applications, implementing SSL offload, persistence profiles, traffic steering, health monitors, and iRules for custom redirection.
- Automated repetitive tasks (VLAN provisioning, interface configs, ACL updates, firewall pushes) using **Python scripts, Ansible playbooks, and REST APIs**, reducing manual workload by 40%.
- Deployed **Zscaler ZIA/ZPA** integrating GRE/IPsec tunnels with branch routers, Azure AD for authentication, and custom access policies for SaaS applications.
- Managed **Infoblox IPAM, DNS, DHCP** to enforce IP allocation, reduce conflicts, and integrate with AD for automated provisioning.
- Implemented **advanced NAC** using Cisco ISE (802.1X, MAB, posture, profiling), securing endpoints with dynamic VLAN assignment and RADIUS/TACACS+.
- Upgraded wireless infrastructure with **Cisco Catalyst 9800 WLCs and Wi-Fi 6 APs**, enabling WPA3, EAP-TLS, and secure onboarding via certificate-based authentication.

- Applied **QoS policies** to prioritize VoIP, video conferencing, and trading applications over MPLS and SD-WAN links, reducing jitter and latency.
- Configured **VPN solutions**: DMVPN for branches, IPsec site-to-site tunnels, SSL VPN for remote users, and AnyConnect with MFA for hybrid workforce.
- Monitored and troubleshot network health using **ThousandEyes, Splunk, SolarWinds, NetFlow, Zabbix, and SNMP traps**, ensuring high availability.
- Produced **HLD/LLD documentation, network diagrams, migration runbooks, structured cabling layouts**, ensuring audit readiness.
- Worked closely with **DevOps/SecOps** to integrate network automation into CI/CD pipelines, validating configs with linting and compliance scans.
- Troubleshot **BGP route leaks, asymmetric routing, multicast issues (PIM, IGMP), MTU fragmentation, and firewall throughput bottlenecks**.
- Supported **cloud networking**: AWS Direct Connect, Azure ExpressRoute, GCP VPN Interconnects, implementing hybrid routing and DR strategies.
- Collaborated with **application, DBA, and InfoSec teams** to secure workloads while ensuring compliance with **SOX, PCI-DSS** standards.

Environment: Cisco Viptela SD-WAN, Cisco ACI, BGP, MPLS L3VPN, VRF, OSPF, Palo Alto NGFW, Panorama, Expedition, Check Point firewall, Cisco ASA, F5 LTM, GTM, iRules, ThousandEyes, Splunk, SolarWinds, NetFlow, Cisco ISE NAC, 802.1X, Wi-Fi 6, WPA3, Zscaler ZIA/ZPA, Infoblox IPAM, Python, Ansible, REST APIs, AWS Direct Connect, Azure ExpressRoute, GCP Interconnect, PCI-DSS, SOX Compliance.

Junior Network Engineer

ICICI BANK MAY 2020-JULY 2023 – Hyderabad, India.

Responsibilities:

- Assisted in configuring and troubleshooting **LAN/WAN technologies** such as VLANs, STP/RSTP/MSTP, EtherChannel, trunking, and inter-VLAN routing on Cisco Catalyst switches, ensuring loop-free, resilient designs.
- Configured **entry-level routing protocols** (OSPF, EIGRP, RIP, static routing) on Cisco ISR and Catalyst devices, supporting branch connectivity and route convergence.
- Provided **firewall administration support** for Cisco ASA, Fortinet, and Check Point firewalls, applying ACL updates, NAT/PAT rules, IPsec VPNs, SSL VPN configurations, and packet filter policies.
- Performed **racking, stacking, patching, and structured cabling** (Cat6, Cat6a, OM3/OM4 fiber, LC/SC connectors) for routers, switches, firewalls, and APs, aligning with TIA/EIA standards.
- Assisted in deploying **wireless LANs** using Cisco WLC 5508 and Aruba controllers, validated AP placement and coverage using **Ekahau, AirMagnet, and Cisco Prime Infrastructure**.
- Monitored enterprise network traffic with **SolarWinds NPM, PRTG, Nagios, and Wireshark packet captures**, escalating performance issues and tracking MTTR.
- Conducted **firmware upgrades, IOS patching, and password recovery** on Cisco IOS, IOS XE, and NX-OS devices during maintenance windows.
- Supported **VoIP infrastructure** (Cisco CME, CUCM), configuring dial-peers, SIP/H.323 protocols, and validating call flows with packet tracing.
- Updated and maintained **network documentation**: Visio topology diagrams, rack elevation maps, structured cabling layouts, IP address inventory, and CMDB entries in ServiceNow.
- Participated in **on-call rotation**, troubleshooting WAN outages, DNS resolution issues, VPN disconnects, and DHCP misconfigurations.
- Assisted in **cloud networking basics**: created AWS VPCs with subnets, route tables, security groups, NAT gateways; configured Azure VNets and NSGs for sandbox environments.

- Conducted **Layer 1/2/3 troubleshooting** including duplex mismatches, spanning-tree loops, OSPF adjacency failures, MTU mismatches, and packet loss analysis.
- Implemented **AAA authentication** (TACACS+, RADIUS) for network device login control, enhancing security posture.
- Supported **identity-based access** projects by integrating basic Cisco ISE profiles for lab testbeds.
- Shadowed senior engineers on **BGP/MPLS labs**, gaining early exposure to enterprise routing strategies.

Environment: VLANs, STP, EtherChannel, OSPF, EIGRP, RIP, ASA firewall, Fortinet firewall, Check Point firewall, NAT, PAT, IPsec VPN, SSL VPN, VoIP, CME, CUCM, Cisco WLC, Aruba WLAN, Ekahau, AirMagnet, SolarWinds, PRTG, Wireshark, ServiceNow CMDB, AWS VPC, Azure VNet, Structured Cabling.

Certification:

- CCNA - Cisco Certified Network Associate
- CCNP - Cisco Certified Network Professional Enterprise
- Aviatrix Multi-Cloud Network Associate
- NI-CLAD – Certified LabView Associate Developer

Currently pursuing: CompTIA Security+, Microsoft Azure (basic exposure, pursuing AZ-900 certification)